

Directed Flow of Identified Particles in Au + Au Collisions

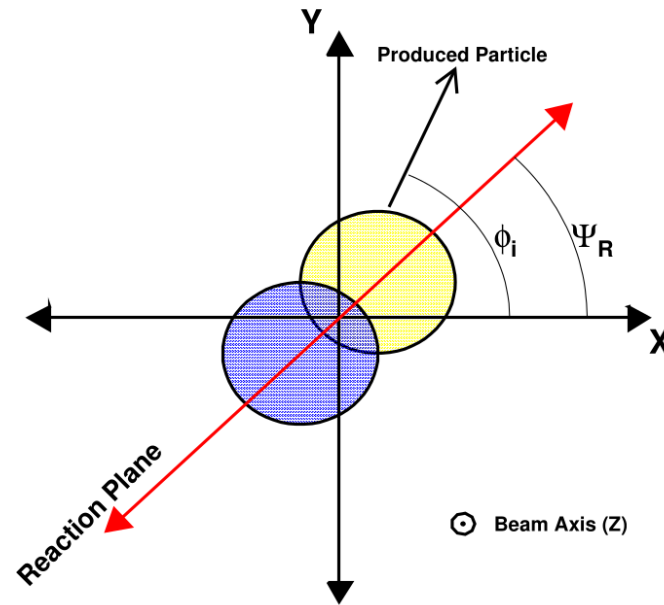


Image reference :

https://www.bnl.gov/userscenter/thesis/past-competitions/2007/files/hamblen_thesis.pdf

Motivation

why v_1 ?

- **Directed flow v_1** measures the collective **sideward motion** of particles
- Initial geometry $\rightarrow$ sideward pressure gradient $\rightarrow v_1$
- Baryon stopping $\rightarrow$ different v_1 for p and p^-
- equation of state $\rightarrow$ pressure response $\rightarrow dv_1/dy$

Contents

- Some basics, what is v1?
- How to measure ?
- Some experimental results and interpretation

Fourier Expansion

$$\frac{dN}{d\phi} = \frac{N}{2\pi} \left[1 + 2 \sum_{n=1}^{\infty} v_n \cos n(\phi - \Psi_n) \right]$$

$$\frac{dN}{dy p_T dp_T d\phi} = \frac{1}{2\pi} \frac{dN}{dy p_T dp_T} \left[1 + 2 \sum_{n=1}^{\infty} v_n(y, p_T) \cos n(\phi - \Psi_n) \right]$$

Why this form ?

$dN/d\phi$ is a periodic function and can be expanded in Fourier series.

Starting from General
Fourier expansion

$$\frac{dN}{d\phi} = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\phi) + b_n \sin(n\phi)]$$

$$\cos[n(\phi - \Psi_n)] = \cos(n\phi) \cos(n\Psi_n) + \sin(n\phi) \sin(n\Psi_n)$$

$$a_n \cos(n\phi) + b_n \sin(n\phi) = A_n \cos[n(\phi - \Psi_n)]$$

$$\frac{dN}{d\phi} = a_0 + \sum_{n=1}^{\infty} A_n \cos[n(\phi - \Psi_n)]$$

Normalization

$$N = \int_0^{2\pi} \frac{dN}{d\phi} d\phi = a_0 \int_0^{2\pi} \left[1 + 2 \sum_{n=1}^{\infty} v_n \cos n(\phi - \Psi_n) \right] d\phi$$

$$N = 2\pi a_0 \quad \Rightarrow \quad \boxed{a_0 = \frac{N}{2\pi}}$$

Look at this average

$$\langle \cos[m(\phi - \Psi_m)] \rangle = \frac{1}{N} \int_0^{2\pi} \left[a_0 + \sum_{n=1}^{\infty} A_n \cos[n(\phi - \Psi_n)] \right] \cos[m(\phi - \Psi_m)] d\phi$$

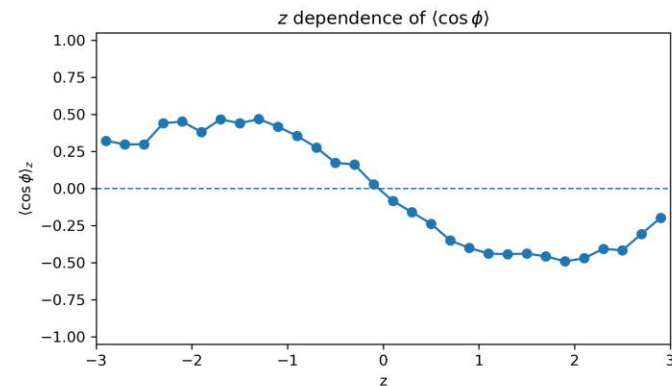
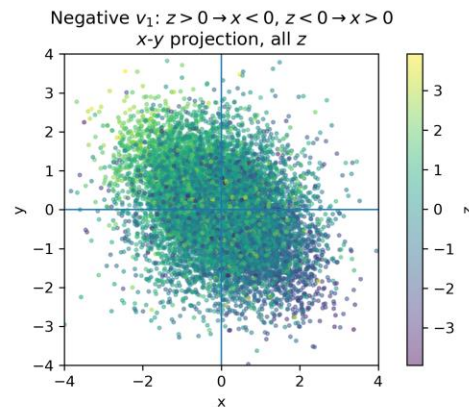
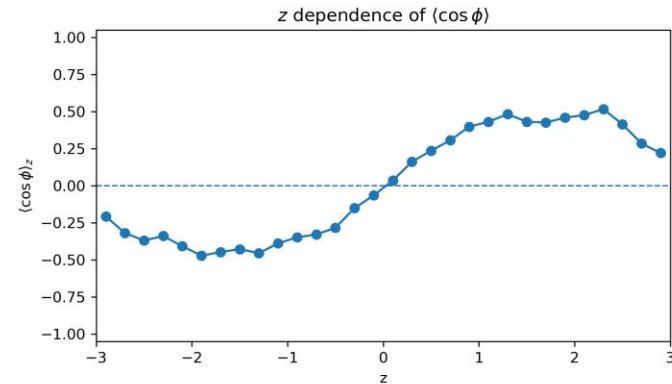
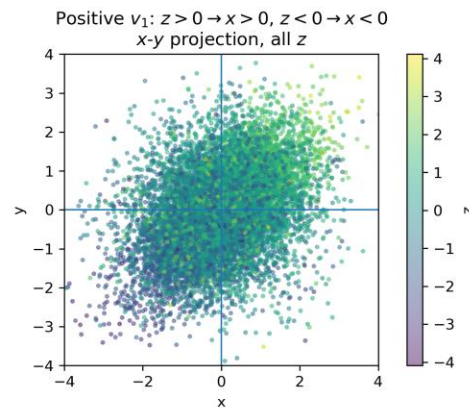
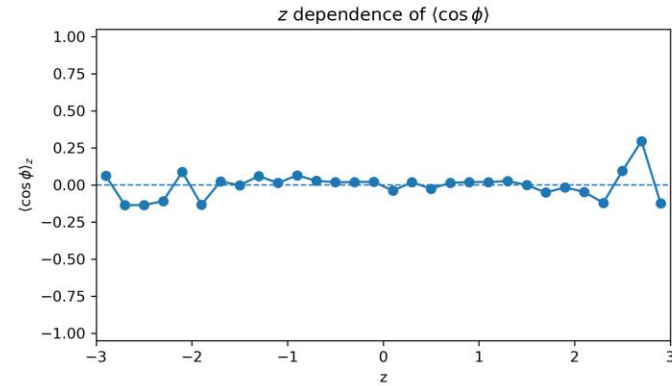
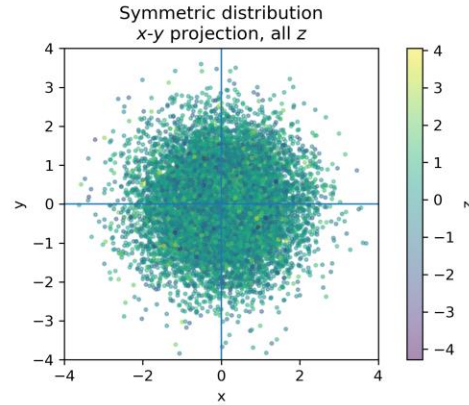
$$\int_0^{2\pi} \cos[n(\phi - \Psi_n)] \cos[m(\phi - \Psi_m)] d\phi = \pi \delta_{mn}$$

$$\langle \cos[m(\phi - \Psi_m)] \rangle = \frac{A_m \pi}{N} = \frac{A_m}{2a_0}$$

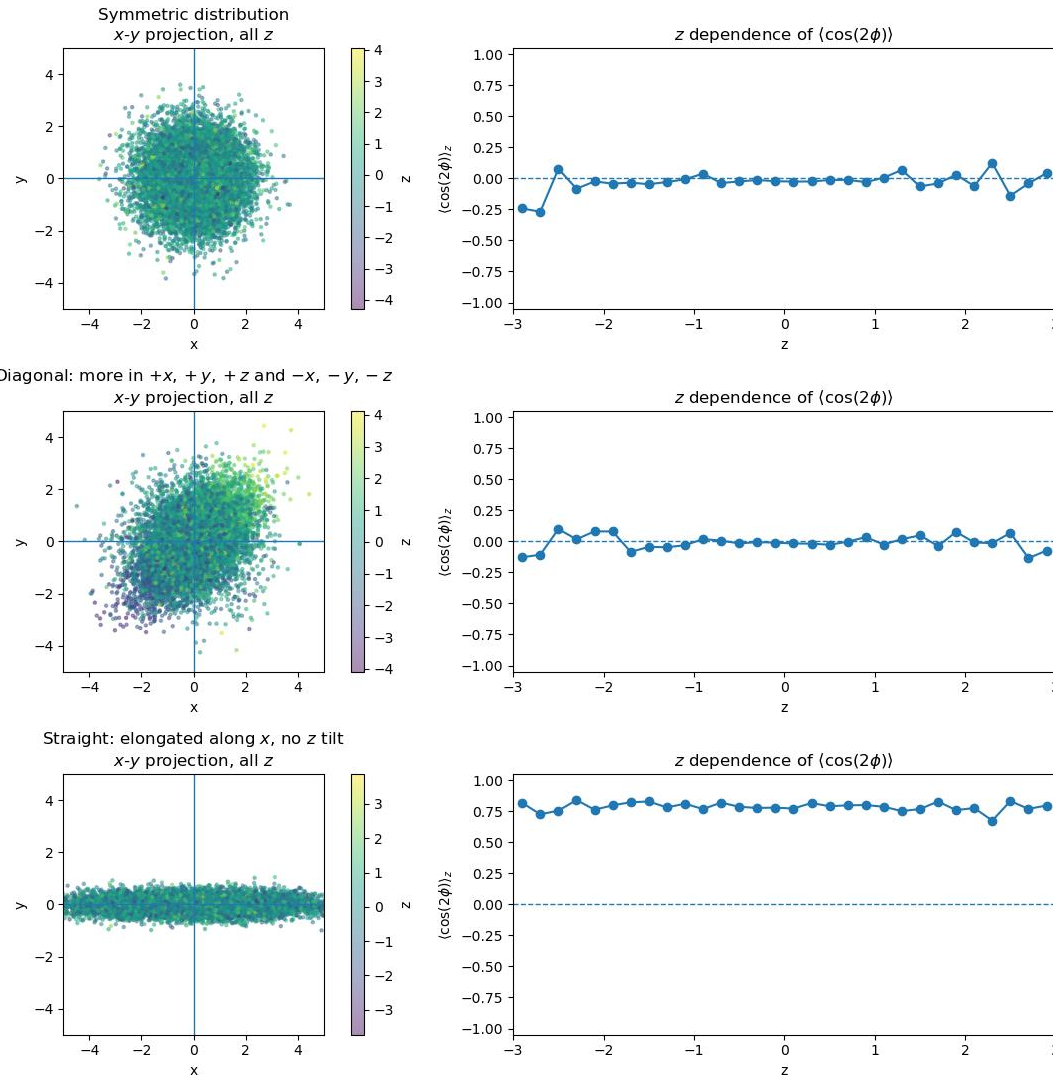
$$A_n = 2a_0 v_n$$

$$\frac{dN}{d\phi} = a_0 \left[1 + 2 \sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_n)] \right]$$

Some illustration

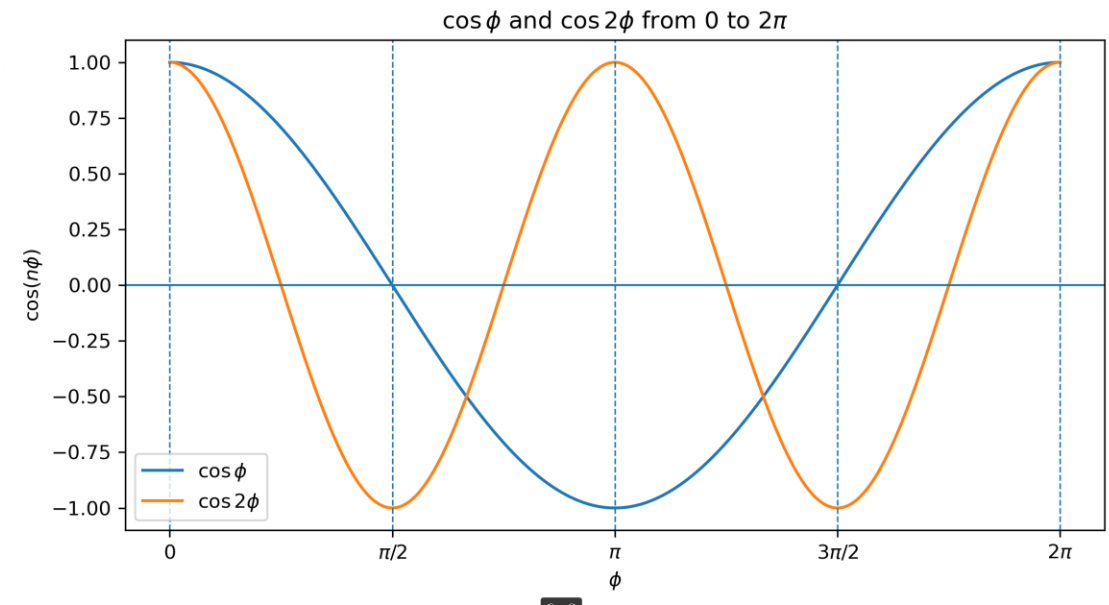


V_2



Note: Just for illustration

- 1) The initial ellipse is assumed to be straight which results in this final expansion.
- 2) More particles in 0 and π , positive contribution to v_2



How to measure v_1 ?

- The reaction plane is not directly known -> estimated reaction plane -> **event plane** is used.
- The event plane is reconstructed from the n th-order event flow vector: Q_n
- The index n refers to the n th component of the Fourier series.

$$\vec{Q}_n = \{X_n, Y_n\}$$

$$X_n = Q_n \cos(n\Psi_n) = \sum_i w_i \cos(n\phi_i)$$

$$Y_n = Q_n \sin(n\Psi_n) = \sum_i w_i \sin(n\phi_i)$$

- ϕ_i is the azimuthal angle and w_i is the weight assigned to the i th particle.

Event plane Resolution

$$\Psi_n = \frac{1}{n} \tan^{-1} \left(\frac{Y_n}{X_n} \right)$$

Previous talk

$$\Psi_n = \frac{1}{n} \tan^{-1} \left(\frac{\sum_i w_i \sin(n\phi_i)}{\sum_i w_i \cos(n\phi_i)} \right)$$

$$\langle \cos(n(\Psi_n^a - \Psi_R)) \rangle = \langle \cos(n(\Psi_n^b - \Psi_R)) \rangle = \sqrt{\langle \cos(n(\Psi_n^a - \Psi_n^b)) \rangle}.$$

$$0 \leq \Psi_n \leq \frac{2\pi}{n}$$

$$v_n = \frac{v_n^{obs}}{\sqrt{\langle \cos n(\Psi_n^a - \Psi_n^b) \rangle}}.$$

$$v_n^{obs} = \langle \cos n(\phi_i - \Psi_n) \rangle$$

$$v_n = \frac{v_n^{obs}}{\langle \cos n(\Psi_n - \Psi_R) \rangle}$$

Some experimental results

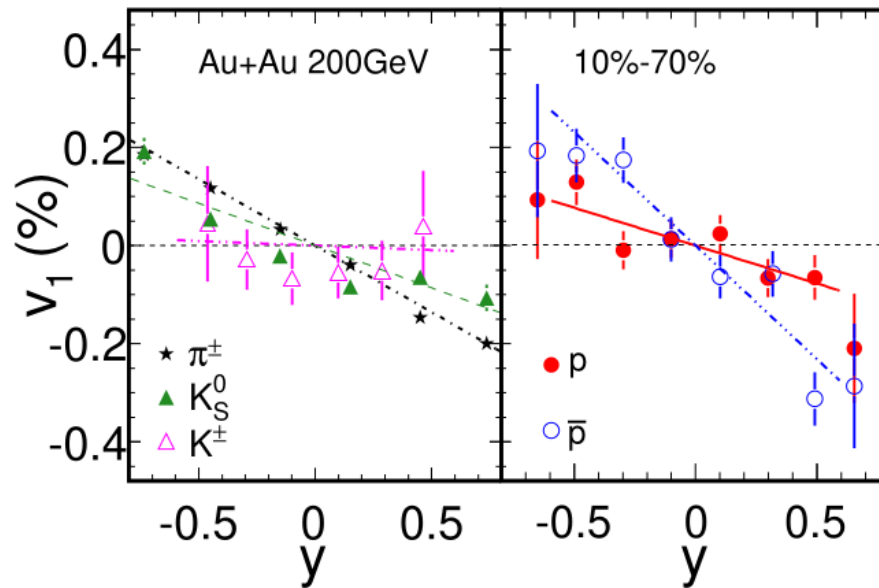


FIG. 1: v_1 for $\pi^\pm$, $K^\pm$, K_S^0 (left panel), p and $\bar{p}$ (right panel) as a function of rapidity for 10-70% Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The lines present the linear fit to the $\pi^\pm$, $K^\pm$, K_S^0 , p and $\bar{p}$'s $v_1(y)$ respectively. Data points around $y = 0.29$ are slightly shifted horizontally to avoid overlapping.

- The **$v_1(y)$ slope** -> **negative** at mid-rapidity
- protons flatter shape than anti-protons.

Centrality dependence

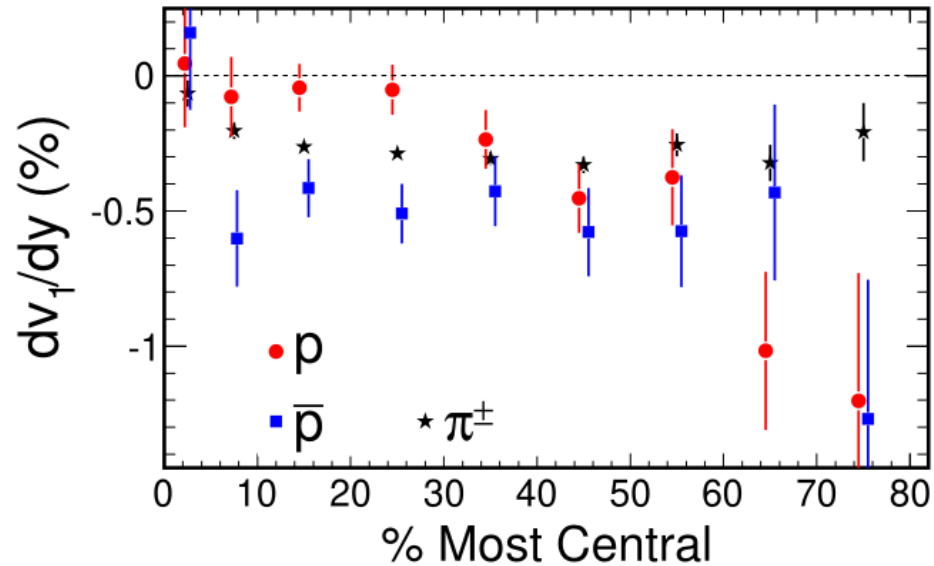


FIG. 3: Charged pions (solid stars), protons (solid circles) and antiprotons (solid squares) $v_1(y)$ slope (dv_1/dy) at midrapidity as a function of centrality for Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV.

- 1) Centrality vs slope of v_1 near mid rapidity
- 2) $v_1(y)$ slope $\rightarrow$ zero $\rightarrow$ most central collisions
- 3) P and P^- $\rightarrow$ consistent in 30-80% and diverge in 5-30%
- 4) Tilted-source $\rightarrow$ but not explain why p and p^- slopes are different.

Identified particle v1

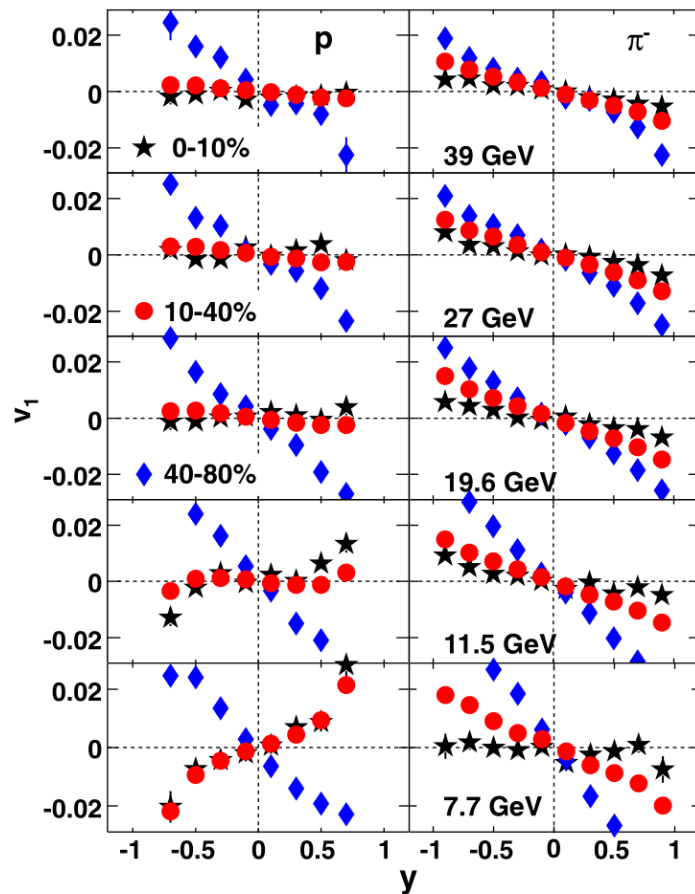


FIG. 1 (color online). Directed flow for protons and π^- versus rapidity for central (0%–10%), intermediate-centrality (10%–40%), and peripheral (40%–80%) Au+Au collisions at $\sqrt{s_{NN}} = 39, 27, 19.6, 11.5$, and 7.7 GeV. The plotted errors are statistical only, and systematic uncertainties are explained in the text.

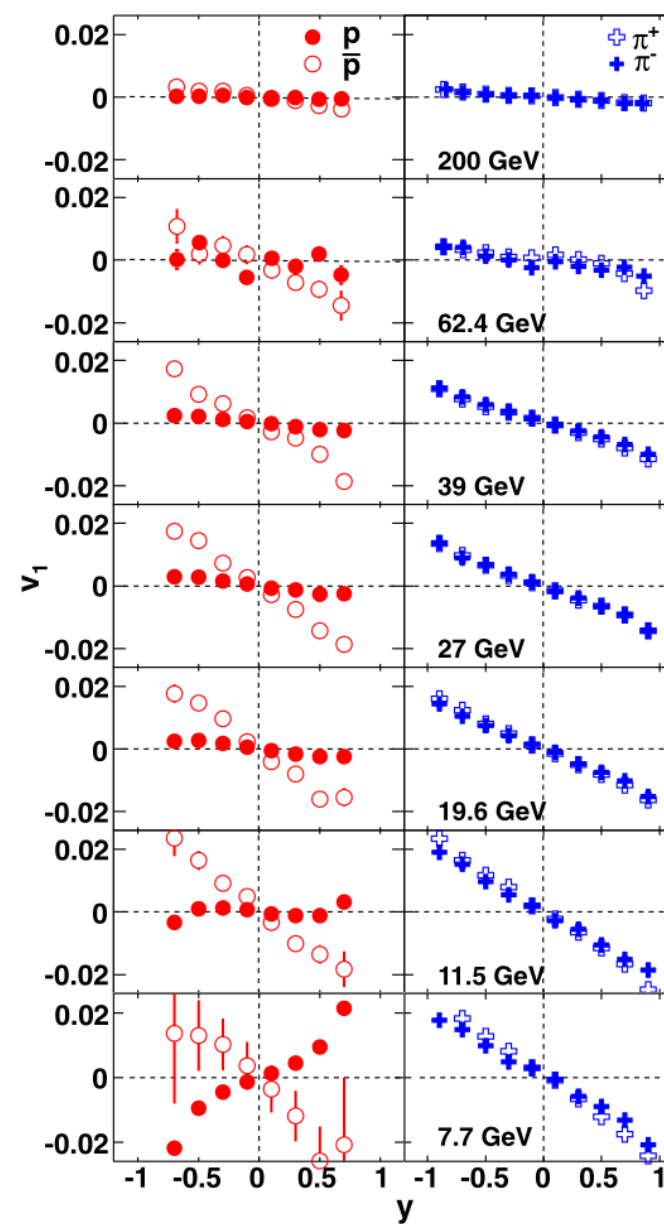


FIG. 2 (color online). Proton and antiproton $v_1(y)$ (left panels) and $\pi^\pm$ $v_1(y)$ (right panels) for intermediate-centrality (10%–40%) Au+Au collisions at 200, 62.4, 39, 27, 19.6, 11.5, and 7.7 GeV. The plotted errors are statistical only.

Energy dependence

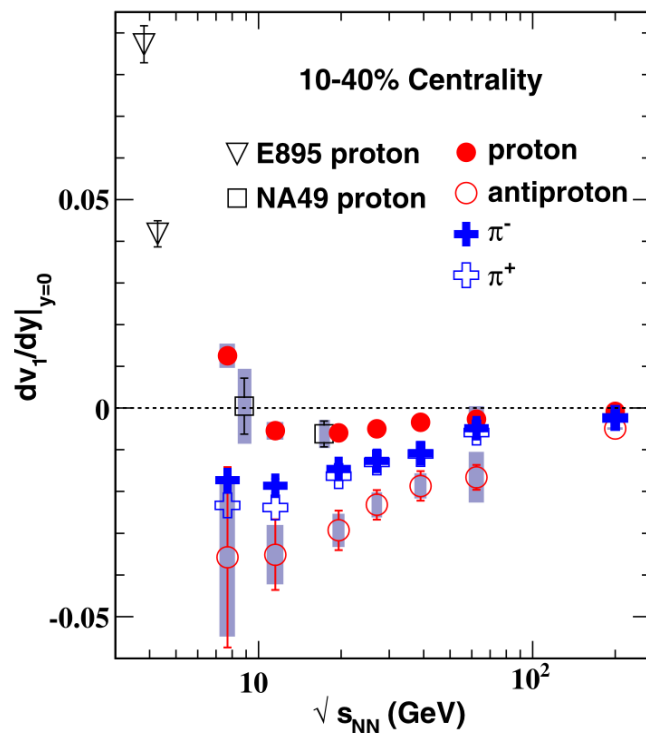


FIG. 3 (color online). Directed flow slope (dv_1/dy) near midrapidity versus beam energy for intermediate-centrality Au+Au collisions. The slopes for protons, antiprotons, and $\pi^\pm$

1) Proton slope sign changes at low energy.

2) pions $\rightarrow$ consistent

3) anti proton and pions have increasing negative slope with decrease in beam energy.

Reference

- https://www.bnl.gov/userscenter/thesis/past-competitions/2007/files/hamblen_thesis.pdf
- <https://arxiv.org/pdf/1112.3930>
- <https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.112.162301>